

Overview of Text-to-Speech Systems

K Sri Rama Murty

IIT Hyderabad

`ksrm@ee.iith.ac.in`

Why Speech?

- Natural
 - Most natural form of communication among humans
 - Does not require any special training

Why Speech?

- Natural
 - Most natural form of communication among humans
 - Does not require any special training
- Flexible
 - Hands/Eyes free communication
 - Natural interaction while performing other tasks - Google Maps
 - Radio survived the era of HD television!
 - Assistive technologies for the blind

Why Speech?

- Natural
 - Most natural form of communication among humans
 - Does not require any special training
- Flexible
 - Hands/Eyes free communication
 - Natural interaction while performing other tasks - Google Maps
 - Radio survived the era of HD television!
 - Assistive technologies for the blind
- Efficient
 - Conveys a lot more information than mere text content

Why Speech?

- Natural
 - Most natural form of communication among humans
 - Does not require any special training
- Flexible
 - Hands/Eyes free communication
 - Natural interaction while performing other tasks - Google Maps
 - Radio survived the era of HD television!
 - Assistive technologies for the blind
- Efficient
 - Conveys a lot more information than mere text content
- Economical
 - Inexpensive transmission and reception of information
 - Voice communication is the reason for the success of mobile phones

Information in Speech Signal

- Linguistic Information
 - Information that can be represented by a set of discrete symbols
 - Conveys textual message in the speech signal
 - Language recognition, speech recognition, search for specific words

Information in Speech Signal

- Linguistic Information
 - Information that can be represented by a set of discrete symbols
 - Conveys textual message in the speech signal
 - Language recognition, speech recognition, search for specific words
- Paralinguistic Information
 - Supplemental information that is not inferable from written text
 - Added by speaker to modify & supplement the linguistic information
 - Intention, attitude, emphasis, speaking style, etc.
 - It can be consciously controlled by the speaker

Information in Speech Signal

- Linguistic Information
 - Information that can be represented by a set of discrete symbols
 - Conveys textual message in the speech signal
 - Language recognition, speech recognition, search for specific words
- Paralinguistic Information
 - Supplemental information that is not inferable from written text
 - Added by speaker to modify & supplement the linguistic information
 - Intention, attitude, emphasis, speaking style, etc.
 - It can be consciously controlled by the speaker
- Nonlinguistic Information
 - Factors that cannot, generally, be controlled by the speaker
 - Gender, speaker identity, age, physical build, emotional state, health, idiosyncrasy (filler words, mother-tongue), etc.

Information in Speech Signal

- Linguistic Information
 - Information that can be represented by a set of discrete symbols
 - Conveys textual message in the speech signal
 - Language recognition, speech recognition, search for specific words
- Paralinguistic Information
 - Supplemental information that is not inferable from written text
 - Added by speaker to modify & supplement the linguistic information
 - Intention, attitude, emphasis, speaking style, etc.
 - It can be consciously controlled by the speaker
- Nonlinguistic Information
 - Factors that cannot, generally, be controlled by the speaker
 - Gender, speaker identity, age, physical build, emotional state, health, idiosyncrasy (filler words, mother-tongue), etc.
- Background Information - Acoustic environment around the speaker

Probabilistic Approach towards Speech Systems

- Speech signal contains several sources of variability

W - Textual message S - Speaker identity

L - Language A - Accent

E - Emotional state B - Ambient background

Probabilistic Approach towards Speech Systems

- Speech signal contains several sources of variability
 - W - Textual message S - Speaker identity
 - L - Language A - Accent
 - E - Emotional state B - Ambient background
- Observed signal $o[n]$ admits a joint distribution with these factors

$$p(w, s, l, a, e, b, \mathbf{o})$$

Probabilistic Approach towards Speech Systems

- Speech signal contains several sources of variability
 - W - Textual message S - Speaker identity
 - L - Language A - Accent
 - E - Emotional state B - Ambient background
- Observed signal $o[n]$ admits a joint distribution with these factors

$$p(w, s, l, a, e, b, o)$$

- Speech recognition - Given the speech, analyze the individual factors
 - Speech recognition - Marginalize all other factors except text W

$$p(w/o) = \sum_{S,L,E,B,A} p(w, s, l, a, e, b/o)$$

Probabilistic Approach towards Speech Systems

- Speech signal contains several sources of variability
 - W - Textual message S - Speaker identity
 - L - Language A - Accent
 - E - Emotional state B - Ambient background
- Observed signal $o[n]$ admits a joint distribution with these factors

$$p(w, s, l, a, e, b, o)$$

- Speech recognition - Given the speech, analyze the individual factors
 - Speech recognition - Marginalize all other factors except text W

$$p(w/o) = \sum_{S,L,E,B,A} p(w, s, l, a, e, b/o)$$

- Speech synthesis - Given factors, synthesize the speech signal
 - Sample from conditional density $p(o/w, s, l, a, e, b)$

Speech Synthesis

- Artificial production of human-like speech from written text.
 - Naturalness: Sounds like a human speaker
 - Intelligibility: Clearly conveys the message
 - Expressiveness: Conveys emotions, prosody, and context

Speech Synthesis

- Artificial production of human-like speech from written text.
 - Naturalness: Sounds like a human speaker
 - Intelligibility: Clearly conveys the message
 - Expressiveness: Conveys emotions, prosody, and context
- Applications
 - Virtual assistants (e.g., Siri, Alexa)
 - Accessibility tools (screen readers for visually impaired)
 - Call centers and customer service
 - Language learning and pronunciation aids
 - Audiobook and content narration
 - Automated dubbing of videos

Approaches to Speech Synthesis

- Concatenative Synthesis (Template Matching)
 - Stitches prerecorded speech units (phonemes, syllables, words)
 - Pros: Natural sound quality (if well-designed)
 - Cons: Rigid, hard to scale, limited expressiveness, memory usage

Approaches to Speech Synthesis

- Concatenative Synthesis (Template Matching)
 - Stitches prerecorded speech units (phonemes, syllables, words)
 - Pros: Natural sound quality (if well-designed)
 - Cons: Rigid, hard to scale, limited expressiveness, memory usage
- Statistical Parametric Synthesis (Statistical Approach)
 - Models speech production mechanism with a smaller set of parameters
 - Pros: Compact, flexible, easy to control prosodic parameters
 - Cons: Typically less natural than the concatenative approach

Approaches to Speech Synthesis

- Concatenative Synthesis (Template Matching)
 - Stitches prerecorded speech units (phonemes, syllables, words)
 - Pros: Natural sound quality (if well-designed)
 - Cons: Rigid, hard to scale, limited expressiveness, memory usage
- Statistical Parametric Synthesis (Statistical Approach)
 - Models speech production mechanism with a smaller set of parameters
 - Pros: Compact, flexible, easy to control prosodic parameters
 - Cons: Typically less natural than the concatenative approach
- Neural End-to-End Synthesis (Neural Approach)
 - Learn to directly map text to spectrograms or waveforms
 - Pros: High naturalness and adaptability
 - Cons: Requires large datasets and computing

Concatenative Speech Synthesis

- Generate speech by concatenating **prerecorded speech segments**
- Choice of the segment (unit)
 - Sentence, phrase, word, syllable, phoneme, diphone

Concatenative Speech Synthesis

- Generate speech by concatenating **prerecorded speech segments**
- Choice of the segment (unit)
 - Sentence, phrase, word, syllable, phoneme, diphone
- Methodology: Viterbi-based unit selection over a candidate lattice
 - **Text Analysis:** Convert the input text into a sequence of target units
 - **Unit Selection:** Choose matching audio segments from the database
 - **Concatenation:** Stitch together selected segments with smoothing

Concatenative Speech Synthesis

- Generate speech by concatenating **prerecorded speech segments**
- Choice of the segment (unit)
 - Sentence, phrase, word, syllable, phoneme, diphone
- Methodology: Viterbi-based unit selection over a candidate lattice
 - **Text Analysis:** Convert the input text into a sequence of target units
 - **Unit Selection:** Choose matching audio segments from the database
 - **Concatenation:** Stitch together selected segments with smoothing
- Synthesizes high-quality natural-sounding speech
- Limited flexibility, large storage, joins can introduce discontinuity

Common Segment Types in Concatenative TTS

Segment Type	Description	Pros	Cons
Word/Phrase	Whole recorded words or frequent phrases	Very natural within-domain synthesis	Extremely large inventory, poor generalization

Common Segment Types in Concatenative TTS

Segment Type	Description	Pros	Cons
Word/Phrase	Whole recorded words or frequent phrases	Very natural within-domain synthesis	Extremely large inventory, poor generalization
Syllable	Group of phonemes forming a syllable (e.g., "ka", "tra")	Fewer joins, better prosody	Large database, hard to ensure full coverage

Common Segment Types in Concatenative TTS

Segment Type	Description	Pros	Cons
Word/Phrase	Whole recorded words or frequent phrases	Very natural within-domain synthesis	Extremely large inventory, poor generalization
Syllable	Group of phonemes forming a syllable (e.g., "ka", "tra")	Fewer joins, better prosody	Large database, hard to ensure full coverage
Phoneme	Smallest sound unit (e.g., /p/, /a/)	Fine-grained control, small inventory	Many joins, unnaturalness due to coarticulation mismatch

Common Segment Types in Concatenative TTS

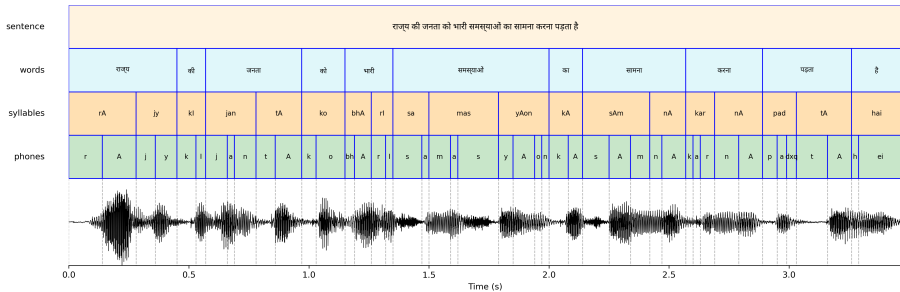
Segment Type	Description	Pros	Cons
Word/Phrase	Whole recorded words or frequent phrases	Very natural within-domain synthesis	Extremely large inventory, poor generalization
Syllable	Group of phonemes forming a syllable (e.g., "ka", "tra")	Fewer joins, better prosody	Large database, hard to ensure full coverage
Phoneme	Smallest sound unit (e.g., /p/, /a/)	Fine-grained control, small inventory	Many joins, unnaturalness due to coarticulation mismatch
Diphone	Transition from middle of one phoneme to middle of the next	Captures coarticulation, balance between quality and flexibility	Still many joins, large database needed

Handling Missing Words

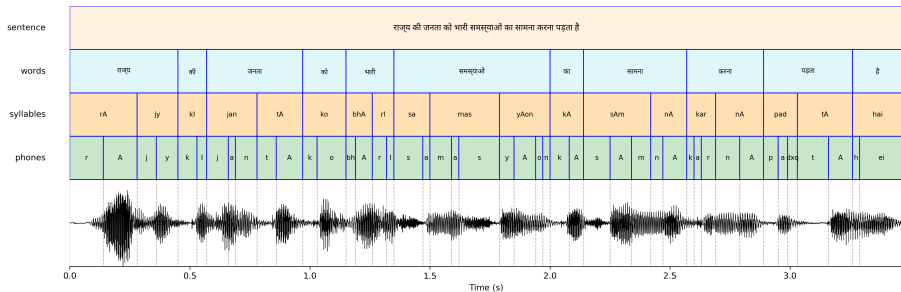
- Target word to be synthesized: **“concatenation”**:
 - Phonemes: /k ɒ n k æ t ɪ n eɪ ʃ ə n/
 - Diphones: k-ɒ | ɒ-n | n-k | k-æ | æ-t | t-ɪ | ɪ-n | n-eɪ | eɪ-ʃ | ʃ-ə | ə-n

Diphone	Donor Word	Transition Context
k-ɒ	“cot”	/k/ into /ɒ/
ɒ-n	“on”	/ɒ/ into /n/
n-k	“ankle”	/n/ into /k/
k-æ	“cat”	/k/ into /æ/
æ-t	“sat”	/æ/ into /t/
t-ɪ	“ticket”	/t/ into /ɪ/
ɪ-n	“in”	/ɪ/ into /n/
n-eɪ	“name”	/n/ into /eɪ/
eɪ-ʃ	“nation”	/eɪ/ into /ʃ/
ʃ-ə	“pressure”	/ʃ/ into /ə/
ə-n	“button”	/ə/ into /n/

Data Preparation

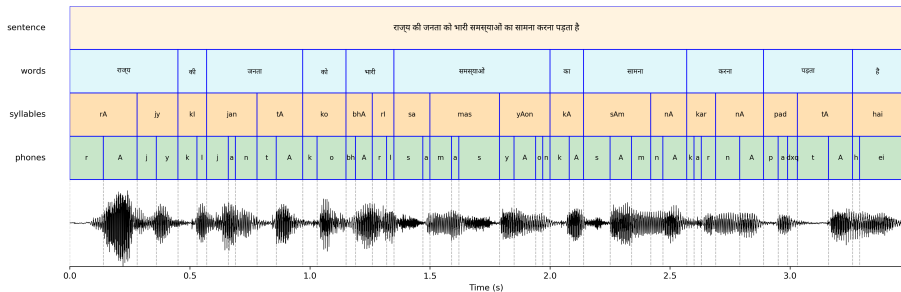


Data Preparation



Phone	Prev. Phone	Next Phone	Phone position	Word Position	Duration	Waveform
/sh/	/a/	/i/	3/6	5/7	65 ms	

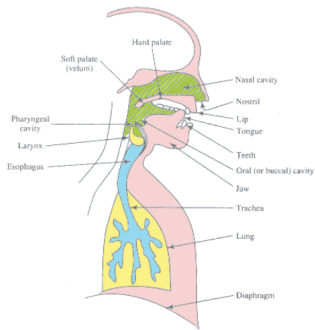
Data Preparation



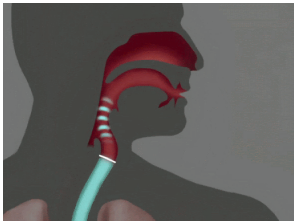
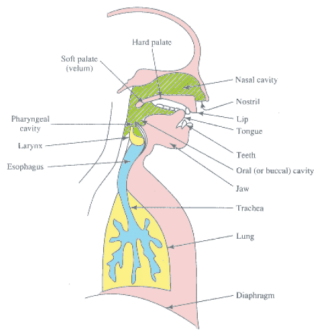
Phone	Prev. Phone	Next Phone	Phone position	Word Position	Duration	Waveform
/sh/	/a/	/i/	3/6	5/7	65 ms	

2h Data - 13h Data

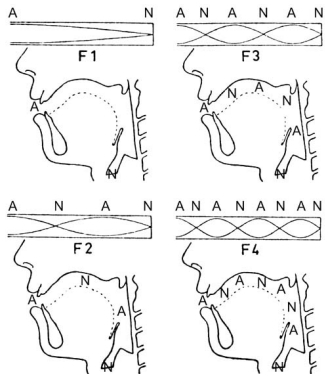
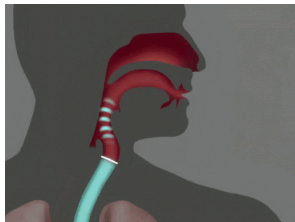
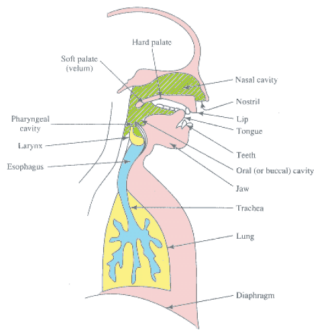
Speech Production Mechanism



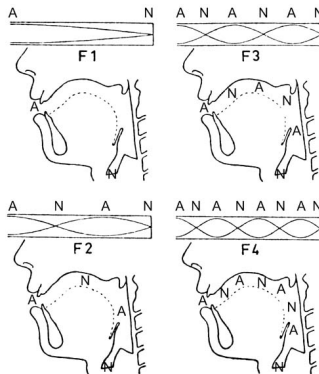
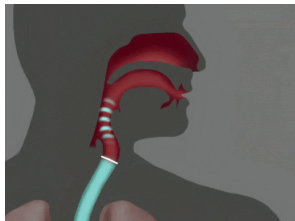
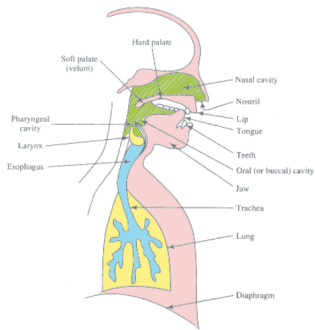
Speech Production Mechanism



Speech Production Mechanism

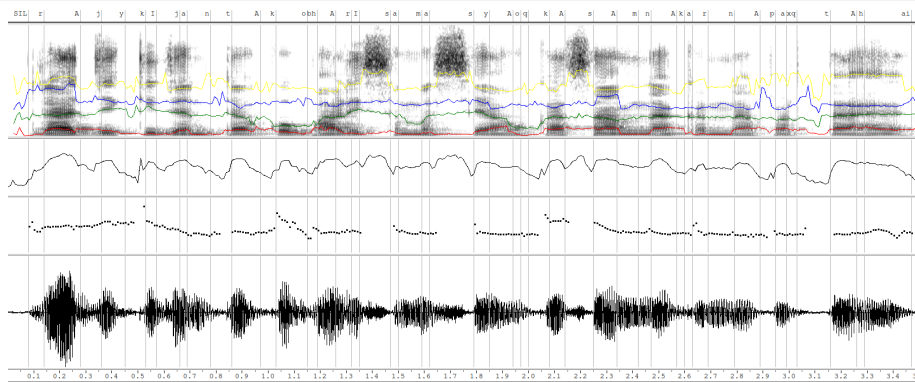


Speech Production Mechanism

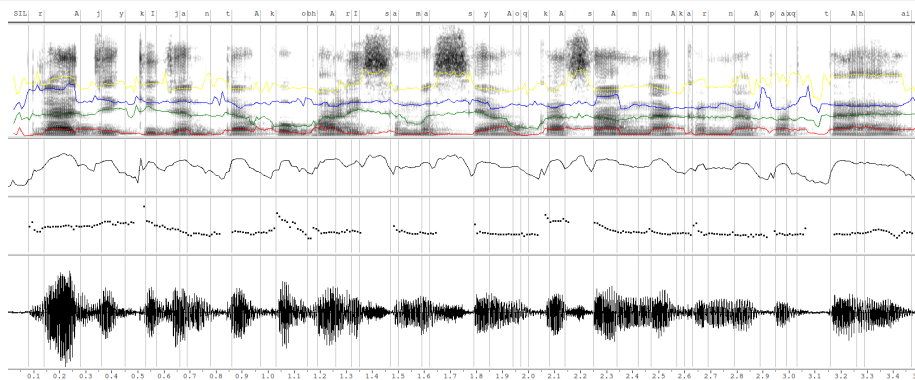


- Important speech parameters
 - Rate of vibration of vocal folds (Pitch)
 - Resonances of vocal tract (Formants)
 - Speaking rate (Duration)

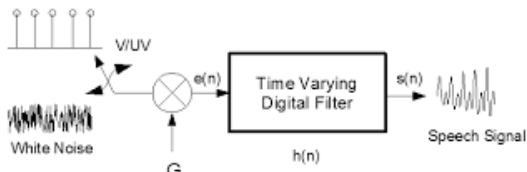
Parameter Contours



Parameter Contours



Periodic Impulses



Parametric Speech Synthesis – An Overview

- Speech is generated by predicting a small set of acoustic parameters
 - Phoneme duration, pitch, energy, and spectral envelope (formants)
 - Waveform is synthesized from these parameters
 - Time-varying all-pole filters driven with periodic/apperiodic excitation

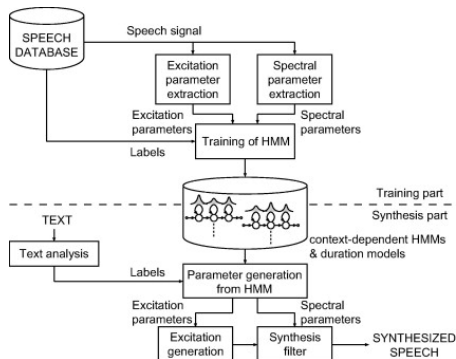
Parametric Speech Synthesis – An Overview

- Speech is generated by predicting a small set of acoustic parameters
 - Phoneme duration, pitch, energy, and spectral envelope (formants)
 - Waveform is synthesized from these parameters
 - Time-varying all-pole filters driven with periodic/apperiodic excitation
- Major steps in parametric synthesis
 - Text Analysis – Extract linguistic/prosodic features
 - Acoustic Modeling – Use HMMs to predict speech parameters
 - Waveform Generation – Synthesize audio using a vocoder
 - E.g., STRAIGHT, WORLD

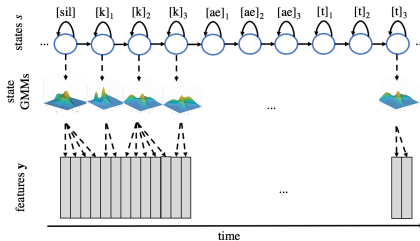
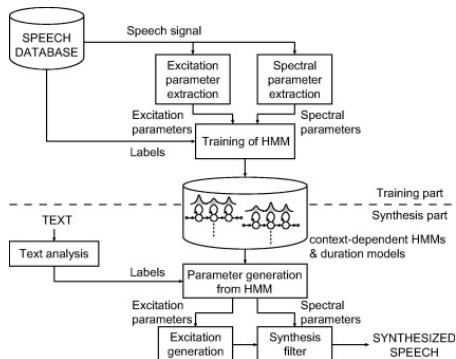
Parametric Speech Synthesis – An Overview

- Speech is generated by predicting a small set of acoustic parameters
 - Phoneme duration, pitch, energy, and spectral envelope (formants)
 - Waveform is synthesized from these parameters
 - Time-varying all-pole filters driven with periodic/apperiodic excitation
- Major steps in parametric synthesis
 - Text Analysis – Extract linguistic/prosodic features
 - Acoustic Modeling – Use HMMs to predict speech parameters
 - Waveform Generation – Synthesize audio using a vocoder
 - E.g., STRAIGHT, WORLD
- Common Approaches:
 - HMM-based Speech Synthesis (HTS)
 - DNN-based acoustic models (e.g., feedforward, RNN)
 - Hybrid approaches (DNN + vocoder refinement)

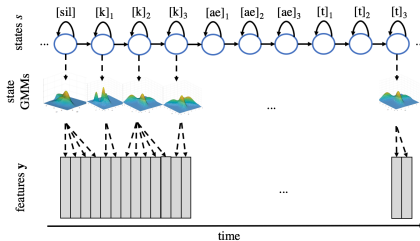
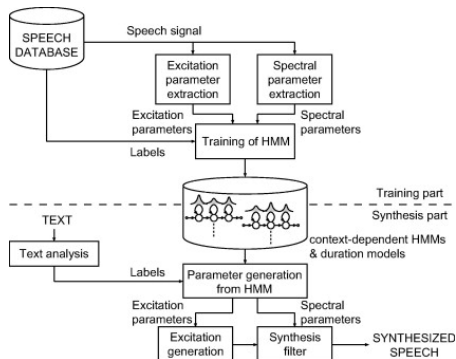
Statistical Parametric Synthesis



Statistical Parametric Synthesis



Statistical Parametric Synthesis



Natural - - Synthesized

Role of Linguistic Knowledge in Traditional TTS Systems

- Traditional TTS rely heavily on explicit linguistic processing
 - Should ensure accurate pronunciation and prosody
 - Pronunciation variation must be resolved before synthesis
 - Text preprocessing involves several steps in traditional TTS systems

Role of Linguistic Knowledge in Traditional TTS Systems

- Traditional TTS rely heavily on explicit linguistic processing
 - Should ensure accurate pronunciation and prosody
 - Pronunciation variation must be resolved before synthesis
 - Text preprocessing involves several steps in traditional TTS systems
- Text Normalization: Expanding abbreviations, numbers, dates
 - 15/10/1931 → fifteenth October nineteen thirty one

Role of Linguistic Knowledge in Traditional TTS Systems

- Traditional TTS rely heavily on explicit linguistic processing
 - Should ensure accurate pronunciation and prosody
 - Pronunciation variation must be resolved before synthesis
 - Text preprocessing involves several steps in traditional TTS systems
- Text Normalization: Expanding abbreviations, numbers, dates
 - 15/10/1931 → fifteenth October nineteen thirty one
- Grapheme-to-Phoneme Conversion (G2P) Rules
 - island → AY L AH N D

Role of Linguistic Knowledge in Traditional TTS Systems

- Traditional TTS rely heavily on explicit linguistic processing
 - Should ensure accurate pronunciation and prosody
 - Pronunciation variation must be resolved before synthesis
 - Text preprocessing involves several steps in traditional TTS systems
- Text Normalization: Expanding abbreviations, numbers, dates
 - 15/10/1931 → fifteenth October nineteen thirty one
- Grapheme-to-Phoneme Conversion (G2P) Rules
 - island → AY L AH N D
- Stress markers to guide prosody and rhythm
 - Voiceless stops /p,t,k/ are aspirated at the beginning of a word (En)

Role of Linguistic Knowledge in Traditional TTS Systems

- Traditional TTS rely heavily on explicit linguistic processing
 - Should ensure accurate pronunciation and prosody
 - Pronunciation variation must be resolved before synthesis
 - Text preprocessing involves several steps in traditional TTS systems
- Text Normalization: Expanding abbreviations, numbers, dates
 - 15/10/1931 → fifteenth October nineteen thirty one
- Grapheme-to-Phoneme Conversion (G2P) Rules
 - island → AY L AH N D
- Stress markers to guide prosody and rhythm
 - Voiceless stops /p,t,k/ are aspirated at the beginning of a word (En)
- Explicit pause tokens to convey the intended meaning
 - Let's eat mangoes vs Let's eat, grandma
 - Incorrect or missing pauses lead to unnatural or unintelligible output.

Strengths and Limitations of Parametric TTS

Aspect	Advantages	Limitations
Flexibility	Easily adaptable to new speakers and styles	May lack expressive control and richness
Model Size	Compact models suitable for embedded devices	Still less natural than neural or concatenative systems
Data Requirement	Requires less data than end-to-end neural systems	Needs clean, well-labeled data and careful tuning
Smoothness	Continuity in prosody and articulation	Can sound overly smoothed or robotic
Vocoder Dependence	Vocoder provides modular control over synthesis	Quality highly dependent on vocoder choice

Neural End-to-End Speech Synthesis

- Learn the entire text-to-waveform mapping using a single model
 - Does not require hand-crafted linguistic feature engineering
 - G2P rules, stress, and phrasing are learned directly from data.
 - Learns to place pauses and intonational cues from training examples
 - Optimizes the entire model jointly, avoiding multiple submodels
 - Uses sequence-to-sequence models with attention or transformers

Neural End-to-End Speech Synthesis

- Learn the entire text-to-waveform mapping using a single model
 - Does not require hand-crafted linguistic feature engineering
 - G2P rules, stress, and phrasing are learned directly from data.
 - Learns to place pauses and intonational cues from training examples
 - Optimizes the entire model jointly, avoiding multiple submodels
 - Uses sequence-to-sequence models with attention or transformers
- Typical architecture of E2E synthesis models
 - **Text Encoder:** Input text $\rightarrow$ latent representations
 - **Acoustic Decoder:** Latent representations $\rightarrow$ Melspectrogram
 - **Vocoder:** Melspectrogram $\rightarrow$ speech waveform

Neural End-to-End Speech Synthesis

- Learn the entire text-to-waveform mapping using a single model
 - Does not require hand-crafted linguistic feature engineering
 - G2P rules, stress, and phrasing are learned directly from data.
 - Learns to place pauses and intonational cues from training examples
 - Optimizes the entire model jointly, avoiding multiple submodels
 - Uses sequence-to-sequence models with attention or transformers
- Typical architecture of E2E synthesis models
 - **Text Encoder:** Input text $\rightarrow$ latent representations
 - **Acoustic Decoder:** Latent representations $\rightarrow$ Melspectrogram
 - **Vocoder:** Melspectrogram $\rightarrow$ speech waveform
- **Popular Models:** Tacotron, ProsodyTTS, FastSpeech, VITS

Text Encoder in Neural TTS

- Maps input characters into a sequence of high-dimensional reps
 - Captures linguistic, positional, and phonological features

Text Encoder in Neural TTS

- Maps input characters into a sequence of high-dimensional reps
 - Captures linguistic, positional, and phonological features
- Embedding layer maps discrete text tokens to continuous vectors
 - Each character is initialized with learnable random vectors

$$a \rightarrow \mathbf{v}_1 \quad b \rightarrow \mathbf{v}_2 \cdots z \rightarrow \mathbf{v}_{26} \quad \mathbf{v}_k \in \mathcal{R}^{256} \quad v_{kl} \sim \mathcal{N}(0, 1)$$

Text Encoder in Neural TTS

- Maps input characters into a sequence of high-dimensional reps
 - Captures linguistic, positional, and phonological features
- Embedding layer maps discrete text tokens to continuous vectors
 - Each character is initialized with learnable random vectors

$$a \rightarrow \mathbf{v}_1 \quad b \rightarrow \mathbf{v}_2 \cdots z \rightarrow \mathbf{v}_{26} \quad \mathbf{v}_k \in \mathcal{R}^{256} \quad v_{kl} \sim \mathcal{N}(0, 1)$$

- Positional encoding adds sequence order information
 - Supplies information about position of a character in the sentence

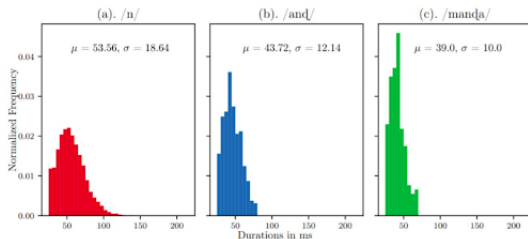
Text Encoder in Neural TTS

- Maps input characters into a sequence of high-dimensional reps
 - Captures linguistic, positional, and phonological features
- Embedding layer maps discrete text tokens to continuous vectors
 - Each character is initialized with learnable random vectors

$$a \rightarrow \mathbf{v}_1 \quad b \rightarrow \mathbf{v}_2 \cdots z \rightarrow \mathbf{v}_{26} \quad \mathbf{v}_k \in \mathcal{R}^{256} \quad v_{kl} \sim \mathcal{N}(0, 1)$$

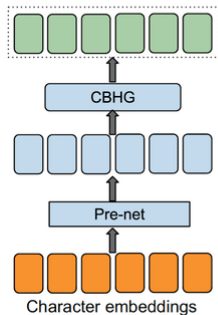
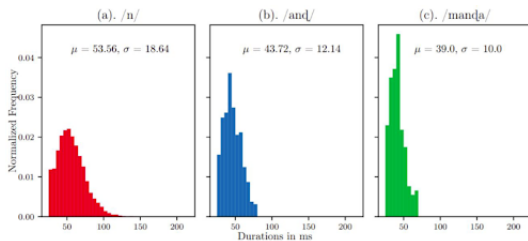
- Positional encoding adds sequence order information
 - Supplies information about position of a character in the sentence
- Encoder network models contextual dependencies across the input.
 - Duration and pronunciation of a character depends on its context
 - Assigns continuous contextual representation the discrete characters
 - With a receptive field of τ , it can capture 2^τ different contexts
 - Implicit context is more expressive than explicit context

Significance of Linguistic Context



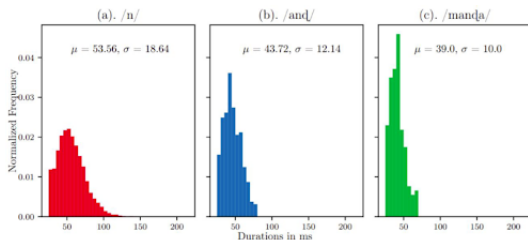
- Longer context reduces variance
- Duration estimation improves

Significance of Linguistic Context

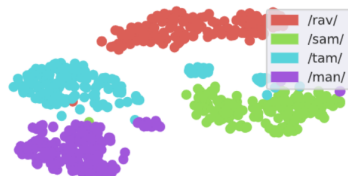
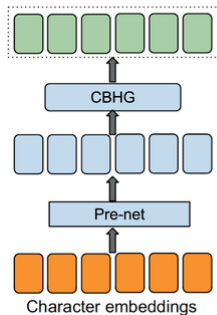


- Longer context reduces variance
- Duration estimation improves

Significance of Linguistic Context



- Longer context reduces variance
- Duration estimation improves



Acoustic Decoder in Neural TTS

- Maps encoded linguistic embeddings to Melspectrogram.
 - Captures prosody, duration, and coarticulation effects across time.

Acoustic Decoder in Neural TTS

- Maps encoded linguistic embeddings to Melspectrogram.
 - Captures prosody, duration, and coarticulation effects across time.
- Core challenge of Rate Mismatch
 - Linguistic input evolves at 10–15 phonemes per second
 - Acoustic output evolves at 200 frames per second
 - Requires careful alignment linguistic and acoustic sequences.

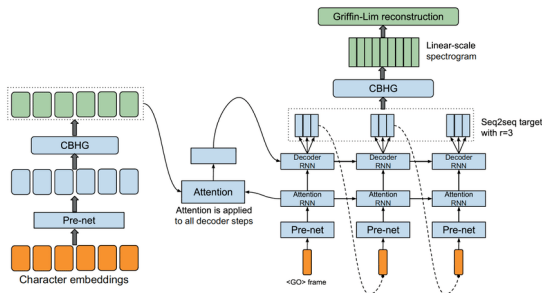
Acoustic Decoder in Neural TTS

- Maps encoded linguistic embeddings to Melspectrogram.
 - Captures prosody, duration, and coarticulation effects across time.
- Core challenge of Rate Mismatch
 - Linguistic input evolves at 10–15 phonemes per second
 - Acoustic output evolves at 200 frames per second
 - Requires careful alignment linguistic and acoustic sequences.
- Explicit duration models
 - Train an explicit duration predictor from ground truth alignment
 - Works even on modest dataset, and offers faster training
- Implicit duration models
 - Uses attention mechanism to learn alignment dynamically.
 - Requires larger dataset and computationally expensive

Acoustic Decoder in Neural TTS

- Maps encoded linguistic embeddings to Melspectrogram.
 - Captures prosody, duration, and coarticulation effects across time.
- Core challenge of Rate Mismatch
 - Linguistic input evolves at 10–15 phonemes per second
 - Acoustic output evolves at 200 frames per second
 - Requires careful alignment linguistic and acoustic sequences.
- Explicit duration models
 - Train an explicit duration predictor from ground truth alignment
 - Works even on modest dataset, and offers faster training
- Implicit duration models
 - Uses attention mechanism to learn alignment dynamically.
 - Requires larger dataset and computationally expensive

Tacotron - Implicit Duration Model

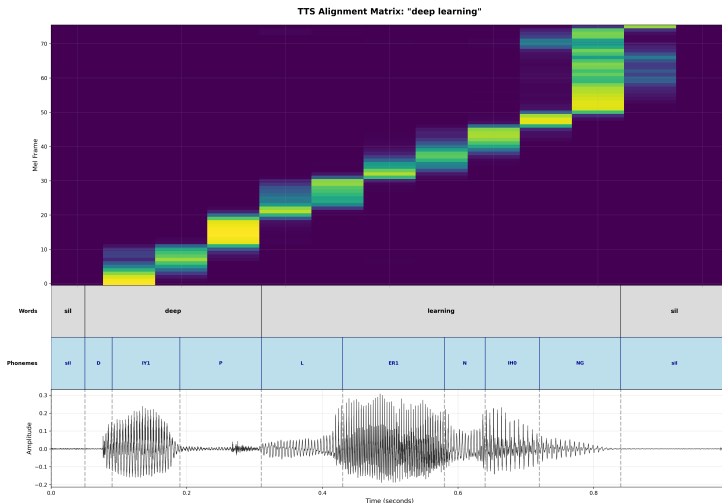


- Acoustic frame is regressed as a linear combination of nonlinearly transformed linguistic embeddings

$$m_t = \sum_{l=1} \alpha_t(l) g(\mathbf{z}_l)$$

- $\alpha_t(l)$ is the attention from t^{th} frame to l^{th} character

Illustration of Implicit Alignment



- Generates high-quality and natural-sounding speech

Issues with Implicit Alignment

- Segment lengths during training influence inference duration
 - Words get repeated for shorter segments
 - Words get deleted for longer segments

Loss Functions in Neural TTS

- Mel-Spectrogram Loss

$$\mathcal{L}_{\text{mel}} = \|\hat{M} - M\|_1 \quad \text{or} \quad \|\hat{M} - M\|_2$$

Where $\hat{M}$ is the predicted mel-spectrogram and M is the target.

- PostNet Loss Refinement loss for post-processed spectrogram
- Duration Loss (explicit models)

$$\mathcal{L}_{\text{dur}} = \|\log(\hat{d} + 1) - \log(d + 1)\|_1$$

- Adversarial / GAN Loss improves naturalness
- Often a weighted combination multiple losses is used

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{mel}} + \lambda_2 \mathcal{L}_{\text{dur}} + \lambda_3 \mathcal{L}_{\text{gan}} + \dots$$

Neural Vocoders in TTS

- Nonlinear neural model for spectrogram inversion
 - Generates waveform from the predicted Melspectrogram
 - Incorporates phase and fine-grained temporal detail directly from data.
 - Responsible for speech naturalness, clarity, and realism.
 - Trained to minimize spectral and perceptual differences from real speech.

Neural Vocoders in TTS

- Nonlinear neural model for spectrogram inversion
 - Generates waveform from the predicted Melspectrogram
 - Incorporates phase and fine-grained temporal detail directly from data.
 - Responsible for speech naturalness, clarity, and realism.
 - Trained to minimize spectral and perceptual differences from real speech.
- Popular Neural Vocoders:
 - **WaveNet**: Autoregressive, high quality but slow inference.
 - **WaveGlow**: Flow-based, faster than WaveNet.
 - **HiFi-GAN**: GAN-based, fast and high-quality; widely used in real-time systems.
 - **Parallel WaveGAN, MelGAN**: Lightweight GAN-based vocoders.

Summary

- Speech synthesis has evolved through three major paradigms:
 - Concatenative TTS: Based on unit selection and waveform stitching; natural but inflexible.
 - Parametric TTS: Uses statistical models (e.g., HMM, DNN) to predict acoustic parameters.
 - Neural End-to-End TTS: Learns the complete mapping from text to waveform with minimal manual intervention.

Summary

- Speech synthesis has evolved through three major paradigms:
 - Concatenative TTS: Based on unit selection and waveform stitching; natural but inflexible.
 - Parametric TTS: Uses statistical models (e.g., HMM, DNN) to predict acoustic parameters.
 - Neural End-to-End TTS: Learns the complete mapping from text to waveform with minimal manual intervention.
- Key Challenges Addressed by Neural TTS:
 - Learns alignment between text and speech implicitly or explicitly.
 - Reduces dependency on linguistic rules (e.g., G2P, stress, pause).
 - Achieves high-quality, natural, and expressive speech synthesis.

Summary

- Speech synthesis has evolved through three major paradigms:
 - Concatenative TTS: Based on unit selection and waveform stitching; natural but inflexible.
 - Parametric TTS: Uses statistical models (e.g., HMM, DNN) to predict acoustic parameters.
 - Neural End-to-End TTS: Learns the complete mapping from text to waveform with minimal manual intervention.
- Key Challenges Addressed by Neural TTS:
 - Learns alignment between text and speech implicitly or explicitly.
 - Reduces dependency on linguistic rules (e.g., G2P, stress, pause).
 - Achieves high-quality, natural, and expressive speech synthesis.
- Ongoing Trends:
 - Non-autoregressive models for fast inference (e.g., FastSpeech)
 - Variational/generative models for expressive speech (e.g., VITS)
 - Multilingual and code-switched TTS for global applications

Ongoing Trends in Speech Synthesis Models

- **Speaker-Specific and Multispeaker Synthesis**

- Learn speaker embeddings for voice cloning and personalized TTS.
- Few-shot or zero-shot speaker adaptation (e.g., using d- or x-vectors)

Ongoing Trends in Speech Synthesis Models

- **Speaker-Specific and Multispeaker Synthesis**

- Learn speaker embeddings for voice cloning and personalized TTS.
- Few-shot or zero-shot speaker adaptation (e.g., using d- or x-vectors)

- **Speech-to-Speech Translation**

- End-to-end models that translate and synthesize speech in another language while preserving speaker traits.
- Leveraging multilingual embeddings and disentangled representations.
- Duration modification and application to automatic dubbing

Ongoing Trends in Speech Synthesis Models

- **Speaker-Specific and Multispeaker Synthesis**

- Learn speaker embeddings for voice cloning and personalized TTS.
- Few-shot or zero-shot speaker adaptation (e.g., using d- or x-vectors)

- **Speech-to-Speech Translation**

- End-to-end models that translate and synthesize speech in another language while preserving speaker traits.
- Leveraging multilingual embeddings and disentangled representations.
- Duration modification and application to automatic dubbing

- **Accent and Style Transfer**

- Modify prosodic and phonetic features to convert accent or expressivity.
- Conditioning TTS on style embeddings or reference audio.

Ongoing Trends in Speech Synthesis Models

- **Speaker-Specific and Multispeaker Synthesis**

- Learn speaker embeddings for voice cloning and personalized TTS.
- Few-shot or zero-shot speaker adaptation (e.g., using d- or x-vectors)

- **Speech-to-Speech Translation**

- End-to-end models that translate and synthesize speech in another language while preserving speaker traits.
- Leveraging multilingual embeddings and disentangled representations.
- Duration modification and application to automatic dubbing

- **Accent and Style Transfer**

- Modify prosodic and phonetic features to convert accent or expressivity.
- Conditioning TTS on style embeddings or reference audio.

- **Unsupervised and Few-Shot Learning**

- Train with limited or no aligned text-speech data.
- Useful for low-resource and endangered languages.

Ongoing Trends in Speech Synthesis Models

- **Speaker-Specific and Multispeaker Synthesis**

- Learn speaker embeddings for voice cloning and personalized TTS.
- Few-shot or zero-shot speaker adaptation (e.g., using d- or x-vectors)

- **Speech-to-Speech Translation**

- End-to-end models that translate and synthesize speech in another language while preserving speaker traits.
- Leveraging multilingual embeddings and disentangled representations.
- Duration modification and application to automatic dubbing

- **Accent and Style Transfer**

- Modify prosodic and phonetic features to convert accent or expressivity.
- Conditioning TTS on style embeddings or reference audio.

- **Unsupervised and Few-Shot Learning**

- Train with limited or no aligned text-speech data.
- Useful for low-resource and endangered languages.

- **Real-Time and Low-Latency Deployment**



Indian Language TTS Datasets

Dataset	Source	Languages	Speaker Coverage	Data Type / Domain
Indic TTS	IIT Madras	22 Major Indian languages	2 M + 2 F per lang 5 h from speaker	Read speech
Syspin Dataset	IISc	Hindi, Bengali, Marathi, Telugu, Kannada, Bhojpuri, Chhattisgarhi, Magahi, Maithili	1 M + 1 F per lang 40 h per speaker	Read speech (Agriculture domain)
RASA	AI4Bharat	22 Major Indian languages	1 M + 1 F per lang 30 h per speaker	Expressive TTS data
Indic Voices - R	AI4Bharat	22 Major Indian languages	10,496 speakers Total 1700 hours	Massive multilingual multi-speaker corpus

Thank You !